

Solving Exponential Equations by Writing a Common Base

Answer Key

Precalculus

Quick Answers

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|------------------|-----------------------------|
| 1. 3 | 6. -5 |
| 2. 4 | 7. 6 |
| 3. $\frac{3}{2}$ | 8. 0 |
| 4. 5 | 9. 20 |
| 5. 3 | 10. $3^{5x+3}, \frac{1}{5}$ |
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Curriculum

Level: Precalculus

HSF-LE.A.4, HSF-BF.B.5 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

Difficulty 1–4 of 5 across 11 tagged checks.

1. $2^{x+1} = 2^{3x-5}$.

Solution: The bases are both 2 and 2^u is one-to-one, so the equation holds exactly when the exponents agree:

$$x + 1 = 3x - 5.$$

Subtract x from both sides: $1 = 2x - 5$. Add 5: $6 = 2x$, so $x = 3$.

Check. Left exponent $3 + 1 = 4$; right exponent $3(3) - 5 = 4$. Both sides are $2^4 = 16$ ✓.

$$\boxed{x = 3}$$

2. $3^{2x} = 3^{x+4}$.

Solution: Same base 3, so equate exponents: $2x = x + 4$, giving $x = 4$.

Check. $2(4) = 8$ and $4 + 4 = 8$; both sides are 3^8 ✓. $\boxed{x = 4}$

3. $9^x = 27$.

Solution: 9 and 27 are both powers of 3, so put the whole equation over base 3: $9 = 3^2$ and $27 = 3^3$, hence

$$(3^2)^x = 3^3 \implies 3^{2x} = 3^3.$$

Equate exponents: $2x = 3$, so $x = \frac{3}{2}$.

Check. $9^{3/2} = (\sqrt{9})^3 = 3^3 = 27$ ✓. A fractional exponent is expected here — nothing says the solution must be an integer. $\boxed{x = \frac{3}{2}}$

4. $5^{3x-1} = 5^{x+9}$.

Solution: Same base 5, so $3x - 1 = x + 9$. Subtract x : $2x - 1 = 9$. Add 1: $2x = 10$, so $x = 5$.

Check. $3(5) - 1 = 14$ and $5 + 9 = 14$; both sides are 5^{14} ✓. Checking the exponents is enough — there is no need to evaluate a 14-digit power. $\boxed{x = 5}$

5. $4^{2x} = 8^{x+1}$.

Solution: $4 = 2^2$ and $8 = 2^3$, so rewrite both sides over base 2:

$$(2^2)^{2x} = (2^3)^{x+1} \implies 2^{4x} = 2^{3x+3}.$$

Note that the outer exponent multiplies the inner one — $(b^m)^n = b^{mn}$ — so the left exponent is $4x$, not $2 + 2x$. Equating exponents gives $4x = 3x + 3$, so $x = 3$.

Check. $4^6 = 4096$ and $8^4 = 4096$ ✓. $\boxed{x = 3}$

6. $(\frac{1}{2})^x = 32$.

Solution: A reciprocal base is a negative exponent: $\frac{1}{2} = 2^{-1}$, so the left side is $(2^{-1})^x = 2^{-x}$. The right side is $32 = 2^5$. Therefore

$$2^{-x} = 2^5 \implies -x = 5 \implies x = -5.$$

Check. $(\frac{1}{2})^{-5} = 2^5 = 32$ ✓. The sign is the whole point: a base below 1 decreases as the exponent grows, so reaching a number above 1 requires a negative exponent. $\boxed{x = -5}$

7. Doubling: $5 \cdot 2^t = 320$.

Solution: The coefficient 5 is not part of the power, so clear it first: divide both sides by 5 to get $2^t = 64$. Now both sides are powers of 2, since $64 = 2^6$:

$$2^t = 2^6 \implies t = 6.$$

Check against the table. The counts run 5, 10, 20, 40, 80, 160, 320, so 320 cells first appear after 6 hours ✓. The table and the algebra are the same statement — the table steps the doubling one hour at a time, and the exponent counts how many doublings have happened. $\boxed{t = 6}$

8. Check the claim: $25^{x-1} = 125^x$.

Solving it independently. $25 = 5^2$ and $125 = 5^3$, so

$$5^{2(x-1)} = 5^{3x} \implies 2x - 2 = 3x \implies x = -2.$$

The student's value is correct.

Confirming numerically. At $x = -2$ the left side is $25^{-3} = \frac{1}{15625}$ and the right side is $125^{-2} = \frac{1}{15625}$. If the claim is true the difference of the two sides must be exactly zero, and it is: $25^{x-1} - 125^x = 0$ at $x = -2$. $\boxed{0}$

Grading: the value asked for is the difference, which must be 0. A student who writes $x = -2$ here has answered the first half; give credit for both parts separately.

9. Decay: $240 \left(\frac{1}{2}\right)^{t/5} = 15$.

Solution: Divide both sides by 240 to isolate the power:

$$\left(\frac{1}{2}\right)^{t/5} = \frac{15}{240} = \frac{1}{16}.$$

Both sides are powers of $\frac{1}{2}$, since $\frac{1}{16} = \left(\frac{1}{2}\right)^4$, so

$$\frac{t}{5} = 4 \implies t = 20 \text{ years.}$$

Half-lives and why it came out whole. $\frac{t}{5} = 4$ says exactly four half-lives have passed. It had to be a whole number because the target mass, 15 mg, is 240 divided by 16, and 16 is a power of 2 — had the target been, say, 20 mg, the ratio $\frac{1}{12}$ is not a power of $\frac{1}{2}$ and the common-base method would not finish the job; that is the case logarithms exist for.

Check. $240 \cdot \frac{1}{16} = 15 \checkmark$. $t = 20$

10. Challenge: $9^x \cdot 27^{x+1} = 81$.

Single power of 3. Rewrite each factor over base 3 and add the exponents:

$$9^x \cdot 27^{x+1} = 3^{2x} \cdot 3^{3x+3} = 3^{2x+3x+3} = 3^{5x+3}.$$

Solving. The right side is $81 = 3^4$, so

$$3^{5x+3} = 3^4 \implies 5x + 3 = 4 \implies x = \frac{1}{5}.$$

Check. $5 \left(\frac{1}{5}\right) + 3 = 4$, so both sides are $3^4 = 81 \checkmark$.

$9^x \cdot 27^{x+1} = 3^{5x+3}, \quad x = \frac{1}{5}$

Watch for: a student who multiplies the exponents instead of adding them gets 3^{6x^2+6x} and a quadratic. The product rule adds; the power rule multiplies.